



Continuum Mechanics

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Chapter 01

Tensor Algebra

Part 4 of 5



Eigenvalues, Eigenvectors, and Principal Invariants

Eigenvalues and Eigenvectors



Definition

By an *eigenpair* for a second-order tensor $S \in \mathcal{V}^2$, we mean a scalar $\lambda \in \mathbb{R}$ and a unit vector $\mathbf{e} \in \mathcal{V}$ satisfying

$$S\mathbf{e} = \lambda\mathbf{e}.$$

Any such λ is called an *eigenvalue* of S , and any such $\mathbf{e}$ is called an *eigenvector* of S .



Proposition

Let $S \in \mathcal{V}^2$ be given. If $\lambda \in \mathbb{R}$ is an eigenvalue of S , then

$$p(\lambda) = 0,$$

where

$$p(\alpha) = \det(S - \alpha I), \quad \alpha \in \mathbb{R}. \quad (1.15)$$

The polynomial p is called the characteristic polynomial of S .



Proposition

Suppose that $S \in \mathcal{V}^2$ is given and $\lambda \in \mathbb{R}$ is an eigenvalue of S . A solution of

$$(S - \lambda I)\mathbf{e} = \mathbf{0} \tag{1.16}$$

yields non-unique eigenvectors of S associated with λ .

Indeed, every nonzero solution of (1.16) can be normalized, and multiplication of an eigenvector by -1 gives another unit eigenvector for the same eigenvalue.



Remark

Here we have decided to ignore complex eigenvalues, not because they are unimportant, but because they are not as useful in continuum mechanics.

The solutions of

$$p(\lambda) = 0$$

could occur in complex-conjugate pairs.



Result 1.8

- ① The eigenvalues of an SPD tensor are strictly positive.
- ② Any two eigenvectors corresponding to distinct eigenvalues of a symmetric tensor are orthogonal.



Proof.

For part (1), let $S \in \mathcal{V}^2$ be SPD and let $(\lambda, \mathbf{e})$ be an eigenpair. Then

$$S\mathbf{e} = \lambda\mathbf{e}.$$

Because $\mathbf{e}$ is a unit vector,

$$\begin{aligned} 0 &< \mathbf{e} \cdot (S\mathbf{e}) \\ &= \mathbf{e} \cdot (\lambda\mathbf{e}) \\ &= \lambda(\mathbf{e} \cdot \mathbf{e}) \\ &= \lambda. \end{aligned}$$

Thus $\lambda > 0$.



Proof (Cont.)

For part (2), suppose that S is symmetric and that $(\lambda, \mathbf{w})$ and $(\mu, \mathbf{u})$ are eigenpairs. Assume $\lambda \neq \mu$. Then

$$\begin{aligned}\lambda(\mathbf{u} \cdot \mathbf{w}) &= \mathbf{u} \cdot (\lambda \mathbf{w}) \\ &= \mathbf{u} \cdot (S\mathbf{w}) \\ &= \mathbf{w} \cdot (S\mathbf{u}) \quad (\text{symmetry}) \\ &= \mathbf{w} \cdot (\mu \mathbf{u}) \\ &= \mu(\mathbf{w} \cdot \mathbf{u}).\end{aligned}$$

Hence

$$(\lambda - \mu)(\mathbf{w} \cdot \mathbf{u}) = 0.$$

Since $\lambda \neq \mu$, cancellation gives

$$\mathbf{w} \cdot \mathbf{u} = 0.$$





Result 1.9 (Spectral Decomposition Theorem)

Let $S \in \mathcal{V}^2$ be symmetric. There is a right-handed orthonormal frame

$$\mathcal{F} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

consisting of eigenvectors of S . The corresponding eigenvalues λ_i , $i = 1, 2, 3$, are the same, up to ordering, for any such frame and form the full set of eigenvalues of S .

The dyadic representation of S in any such frame is

$$S = \sum_{i=1}^3 \lambda_i \mathbf{e}_i \otimes \mathbf{e}_i,$$

and its matrix representation is

$$[S] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}.$$



Result 1.9 (Cont.)

The three eigenvalues are unique up to ordering, even when an eigenspace has dimension greater than one. Within a repeated-eigenvalue eigenspace, the orthonormal eigenvectors need not be unique.

Proof.

A standard result from linear algebra.



Principal Invariants



Definition

Let $S \in \mathcal{V}^2$ be arbitrary. The *principal invariants* of S are the three scalar quantities

$$I_1(S) = \text{tr}(S),$$

$$I_2(S) = \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)],$$

$$I_3(S) = \det(S).$$



Proposition

Let $S \in \mathcal{V}^2$ be a given tensor and $\alpha \in \mathbb{R}$ a scalar. Then

$$\det(S - \alpha I) = -\alpha^3 + I_1(S)\alpha^2 - I_2(S)\alpha + I_3(S) = p(\alpha).$$

Proof.

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame. Compute the characteristic polynomial $p(\alpha) = \det([S] - \alpha[I])$ and compare coefficients with the definitions of $I_1(S)$, $I_2(S)$, and $I_3(S)$. □



Proposition

Let $S \in \mathcal{V}^2$ be symmetric, with eigenvalues $\lambda_1, \lambda_2, \lambda_3$. Then

$$I_1(S) = \lambda_1 + \lambda_2 + \lambda_3,$$

$$I_2(S) = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1,$$

$$I_3(S) = \lambda_1\lambda_2\lambda_3.$$

Proof.

Exercise.





Remark

The quantities l_1 , l_2 , and l_3 are called invariants because they are built from the invariant quantities $\text{tr}(S)$ and $\det(S)$. See Results 1.5 and 1.6.



Result 1.10 (Cayley–Hamilton Theorem)

Let $S \in \mathcal{V}^2$ be arbitrary. Then

$$S^3 - I_1(S)S^2 + I_2(S)S - I_3(S)I = O.$$

Proof.

Exercise.





Special Decompositions



Result 1.11 (The SPD Square Root)

Let $C \in \mathcal{V}^2$ be SPD. There is a unique SPD tensor U such that

$$U^2 = C.$$

In particular,

$$U = \sum_{i=1}^3 \sqrt{\lambda_i} \mathbf{e}_i \otimes \mathbf{e}_i,$$

where $\mathcal{F} = \{\mathbf{e}_i\}$ is any frame of eigenvectors of C and $\lambda_i > 0$ are the corresponding eigenvalues of C . We will write, in this case, $U = \sqrt{C}$.

Proof.

Exercise. □



Result 1.12 (Polar Decomposition)

Let $F \in \mathcal{V}^2$ satisfy

$$\det(F) > 0.$$

Then there exist right and left polar decompositions of F of the forms, respectively,

$$F = RU \quad \text{and} \quad F = VR,$$

where

$$U = \sqrt{F^T F} \quad \text{and} \quad V = \sqrt{F F^T}$$

are both SPD, and R is a rotation.



Proof.

Since $\det(F) > 0$, the tensor F is invertible. Thus

$$F\mathbf{w} \neq \mathbf{0} \quad \text{for every } \mathbf{w} \in \mathcal{V}_* := \mathcal{V} \setminus \{\mathbf{0}\}.$$

Similarly, F^T is invertible and

$$F^T \mathbf{w} \neq \mathbf{0} \quad \text{for every } \mathbf{w} \in \mathcal{V}_*.$$

Therefore,

$$\mathbf{w} \cdot (F^T F \mathbf{w}) = (F \mathbf{w}) \cdot (F \mathbf{w}) > 0,$$

$$\mathbf{w} \cdot (F F^T \mathbf{w}) = (F^T \mathbf{w}) \cdot (F^T \mathbf{w}) > 0,$$

for every $\mathbf{w} \in \mathcal{V}_*$.



Proof (Cont.)

Thus $F^T F$ and FF^T are SPD. Using Result 1.11, define

$$U := \sqrt{F^T F}, \quad V := \sqrt{FF^T}.$$

Next, define

$$R := FU^{-1}.$$

Then $RU = F$. Observe that

$$\begin{aligned} 0 < \det(F) &= \det(RU) \\ &= \det(R) \det(U). \end{aligned}$$

Also, since U is SPD,

$$\det(U) > 0.$$

Therefore $\det(R) > 0$.



Proof (Cont.)

Furthermore,

$$\begin{aligned} R^T R &= U^{-T} F^T F U^{-1} \\ &= U^{-1} U^2 U^{-1} \\ &= I, \end{aligned}$$

because $U^T = U$. Hence R is orthogonal, so

$$|\det(R)| = 1.$$

Together with $\det(R) > 0$, this gives $\det(R) = 1$. Thus R is a rotation, and the right polar decomposition

$$F = RU$$

is proved.



Proof (Cont.)

To obtain the left polar decomposition, define

$$Q := V^{-1}F.$$

The same calculation used for R shows that Q is a rotation. Hence

$$RU = F = VQ.$$

Let

$$S := Q^T VQ.$$

Because V is SPD and Q is orthogonal, S is SPD. Indeed, $S^T = S$, and for $\mathbf{w} \neq \mathbf{0}$,

$$\mathbf{w} \cdot (S\mathbf{w}) = (Q\mathbf{w}) \cdot (VQ\mathbf{w}) > 0.$$



Proof (Cont.)

Using $RU = VQ$,

$$\begin{aligned} RU &= VQ \\ &= QQ^T VQ \\ &= QS, \end{aligned}$$

which implies

$$R = QSU^{-1}.$$

Since R and Q are rotations,

$$R^T R = I, \quad Q^T Q = I.$$

Also, $S^T = S$ and $U^T = U$ because S and U are SPD.



Proof (Cont.)

Using $R = QSU^{-1}$ and the identities from the previous slide,

$$\begin{aligned} I &= R^T R \\ &= (QSU^{-1})^T (QSU^{-1}) \\ &= U^{-T} S^T Q^T Q S U^{-1} \\ &= U^{-1} S^2 U^{-1}. \end{aligned}$$

Multiplying this equality on the left and on the right by U gives

$$U^2 = S^2.$$

Since $U^2 = F^T F$ by definition,

$$S^2 = U^2 = F^T F.$$

Thus S and U are both SPD square roots of $F^T F$.



Proof (Cont.)

By uniqueness of the SPD square root, $U = S$. Hence

$$R = QSU^{-1} = Q.$$

Therefore,

$$RU = F = VR.$$

This gives both polar decompositions with the same rotation tensor R . □



Scalar Product for Second-Order Tensors

Tensor Inner Product



Definition

Let $S, D \in \mathcal{V}^2$. The *tensor inner product* is defined by

$$S : D := \text{tr}(S^T D). \quad (1.17)$$



Proposition

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame. Then the dyadic basis for $\mathcal{V}^2$,

$$\{\mathbf{e}_i \otimes \mathbf{e}_j \mid 1 \leq i, j \leq 3\},$$

is orthonormal with respect to the tensor inner product; that is,

$$(\mathbf{e}_i \otimes \mathbf{e}_j) : (\mathbf{e}_k \otimes \mathbf{e}_\ell) = \delta_{ik} \delta_{j\ell}.$$

Proof.

Exercise. □



Result 1.13

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame. Suppose

$$S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad D = D_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$S : D = S_{ij} D_{ij}.$$

If S is symmetric, then

$$S : D = S : \text{sym}(D) = \text{sym}(S) : \text{sym}(D).$$



Proof.

Using the dyadic expansions and orthonormality of the dyadic basis,

$$\begin{aligned} S : D &= (S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j) : (D_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij} D_{k\ell} (\mathbf{e}_i \otimes \mathbf{e}_j) : (\mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij} D_{k\ell} \delta_{ik} \delta_{j\ell} \\ &= S_{ij} D_{ij}. \end{aligned}$$

Here all repeated indices are dummy indices; the contractions with δ_{ik} and $\delta_{j\ell}$ leave no free index.



Proof (Cont.)

If S is symmetric, then $S_{ij} = S_{ji}$. Therefore,

$$\begin{aligned} S : D &= S_{ij} D_{ij} \\ &= \frac{1}{2} (S_{ij} D_{ij} + S_{ji} D_{ji}) \\ &= \frac{1}{2} S_{ij} (D_{ij} + D_{ji}) \\ &= S_{ij} [\text{sym}(D)]_{ij} \\ &= S : \text{sym}(D). \end{aligned}$$

Since $\text{sym}(S) = S$, it follows that

$$S : D = \text{sym}(S) : \text{sym}(D).$$





Fourth-Order Tensors

Fourth-Order Tensors



Definition

A *fourth-order tensor* is a mapping

$$\mathbf{C} : \mathcal{V}^2 \longrightarrow \mathcal{V}^2$$

that is linear in the sense that

$$\mathbf{C}(S + T) = \mathbf{C}(S) + \mathbf{C}(T), \quad S, T \in \mathcal{V}^2,$$

$$\mathbf{C}(\alpha T) = \alpha \mathbf{C}(T), \quad \alpha \in \mathbb{R}, \quad T \in \mathcal{V}^2.$$

The set of all fourth-order tensors is denoted $\mathcal{V}^4$.

Zero, Identity, and Equality



The *zero fourth-order tensor* $\mathbf{O} \in \mathcal{V}^4$ is characterized by

$$\mathbf{O}S = \mathbf{O}, \quad S \in \mathcal{V}^2.$$

The *identity fourth-order tensor* $\mathbf{I} \in \mathcal{V}^4$ is characterized by

$$\mathbf{I}S = S, \quad S \in \mathcal{V}^2.$$

Two fourth-order tensors $\mathbf{C}, \mathbf{D} \in \mathcal{V}^4$ are called *equal* if and only if

$$\mathbf{C}T = \mathbf{D}T, \quad T \in \mathcal{V}^2.$$



Example

Fix $A \in \mathcal{V}^2$. Define

$$\mathbf{C}(T) = AT, \quad T \in \mathcal{V}^2.$$

Then $\mathbf{C}$ is a fourth-order tensor, because

$$\mathbf{C}(S + T) = A(S + T) = AS + AT = \mathbf{C}(S) + \mathbf{C}(T),$$

$$\mathbf{C}(\alpha T) = A(\alpha T) = \alpha AT = \alpha \mathbf{C}(T).$$



Proposition

Let $\mathbf{C}, \mathbf{D} \in \mathcal{V}^4$ and $\alpha, \beta \in \mathbb{R}$. Define

$$(\alpha \mathbf{C} + \beta \mathbf{D})T := \alpha \mathbf{C}(T) + \beta \mathbf{D}(T), \quad T \in \mathcal{V}^2.$$

Define the product $\mathbf{CD} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ by

$$(\mathbf{CD})(T) := \mathbf{C}(\mathbf{D}(T)), \quad T \in \mathcal{V}^2.$$

Then $\mathcal{V}^4$ is a linear vector space. Furthermore, $\mathbf{CD} \in \mathcal{V}^4$.

Proof.

An easy exercise using the linearity of $\mathbf{C}$ and $\mathbf{D}$. □



Representation in a Coordinate Frame

Components of a Fourth-Order Tensor



Definition

Let $\mathbf{C} \in \mathcal{V}^4$ and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is a right-handed orthonormal frame. The 81 components of $\mathbf{C}$,

$$C_{ijkl} \in \mathbb{R}, \quad 1 \leq i, j, k, \ell \leq 3,$$

are defined by

$$C_{ijkl} = \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell)\mathbf{e}_j).$$



Proposition

Suppose that $\mathbf{C} \in \mathcal{V}^4$ and $U, T \in \mathcal{V}^2$ satisfy

$$U = \mathbf{C}(T).$$

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame and write

$$U = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad T = T_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$U_{ij} = C_{ijkl} T_{kl}. \quad (1.18)$$

Here i, j are free indices, while k, ℓ are dummy indices.



Proof.

We have

$$\begin{aligned} U_{ij} &= \mathbf{e}_i \cdot (U \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) T_{k\ell} \\ &= C_{ijk\ell} T_{k\ell}. \end{aligned}$$

The free indices i, j agree in every line, and k, ℓ are dummy indices. □



Example

Let $A \in \mathcal{V}^2$ be given and define $\mathbf{C} \in \mathcal{V}^4$ by

$$\mathbf{C}(T) = AT.$$

Then

$$\begin{aligned} C_{ijk\ell} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot ((A\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A\mathbf{e}_k (\mathbf{e}_\ell \cdot \mathbf{e}_j)) \\ &= \delta_{\ell j} \mathbf{e}_i \cdot (A\mathbf{e}_k) \\ &= A_{ik} \delta_{\ell j}. \end{aligned}$$



Proposition

Let $\mathbf{C} \in \mathcal{V}^4$. Then its components may equivalently be written

$$C_{ijkl} = (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l).$$



Proof.

We first show that, for any $A \in \mathcal{V}^2$,

$$A_{ij} = \mathbf{e}_i \cdot (A\mathbf{e}_j) = (\mathbf{e}_i \otimes \mathbf{e}_j) : A.$$

Indeed,

$$\begin{aligned}\mathbf{e}_i \otimes \mathbf{e}_j &= \delta_{pi} \delta_{qj} \mathbf{e}_p \otimes \mathbf{e}_q, & (\text{sum on } p, q), \\ (\mathbf{e}_i \otimes \mathbf{e}_j) : A &= \delta_{pi} \delta_{qj} (\mathbf{e}_p \otimes \mathbf{e}_q) : A \\ &= \delta_{pi} \delta_{qj} A_{pq} \\ &= A_{ij},\end{aligned}$$

where Result 1.13 is applied to each dyad $\mathbf{e}_p \otimes \mathbf{e}_q$. Here i, j are free, while p, q are dummy.



Proof (Cont.)

For fixed k, ℓ , define

$$A^{(k\ell)} := \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \in \mathcal{V}^2.$$

Applying the identity from the previous slide to $A^{(k\ell)}$ gives

$$\begin{aligned} C_{ijkl} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A^{(k\ell)} \mathbf{e}_j) \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : A^{(k\ell)} \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell), \end{aligned}$$

which proves the result. □